

ERICK REINALDO FLORES ZAMBRANO

Economist & Data Scientist | AI Engineer | ML Engineer

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PROFESSIONAL SUMMARY

Economist and Data Scientist with dual academic formation (Economics + Data Science & AI), specialized in the intersection of financial analysis, machine learning, and intelligent automation. Demonstrated experience building end-to-end AI solutions: from predictive models in Python (scikit-learn, PyTorch, TensorFlow) to conversational agent architectures with LLMs, API integrations, and MCP (Model Context Protocol) for multi-agent systems. Currently pursuing two simultaneous university degrees — a testament to discipline, self-directed learning, and resilience. Seeking a remote opportunity to contribute and grow.

KEY PROJECTS & PORTFOLIO

FCH-ARX V4 — Original Cryptographic Algorithm [Python, C, NIST FIPS 180-4]

- Designed and implemented a novel hash algorithm from scratch using ARX (Add-Rotate-XOR) operations with mathematically derived constants (Tesla 3-6-9, 26 rounds). Passed SAC-NIST test with 49.95% avalanche (SHA-256 reference: 50.02%). Speed: 185 MB/s in optimized C. Attack complexity: 2^{256} .

AI Job Hunter Bot V4 — Autonomous 24/7 Agent [Python, LinkedIn API, Gmail API, OpenRouter, GitHub Actions]

- Full automation pipeline: scraping 9+ job platforms, real email verification via Hunter.io API, personalized cover letters generated with LLMs, automatic Gmail sending with CV attachment, anti-duplicate memory, and 24/7 deployment on GitHub Actions. This CV was sent using this bot.

Algorithmic Trading with Deep Learning [PyTorch, LSTM, Backtrader, MLflow, Python]

- LSTM model for S&P 500 price forecasting achieving 68% directional accuracy. Complete pipeline: data ingestion, feature engineering, momentum/mean-reversion backtesting, experiment tracking with MLflow.

Multi-Agent System with MCP (Model Context Protocol) [Python, FastAPI, Docker, LLMs, GARCH]

- Distributed agent architecture for automated financial analysis: scraping agent (Alpha Vantage, TradingView), econometric agent (GARCH volatility models), LLM agent for executive report generation.

ETL Data Warehouse — Star Schema [Pentaho, PostgreSQL, SQL, ETL]

- Complete ETL pipeline for an airline Data Warehouse: dimension tables (Aircraft, Route, Flight, Passenger), SQL data cleaning, transformations in Pentaho Data Integration, production-level PostgreSQL integration.

RESTful API for Predictive Models [FastAPI, Docker, Random Forest, GitHub Actions]

- ML inference service in production: credit risk classification (Random Forest), Docker containerization, automatic Swagger documentation, pytest testing, basic CI/CD with GitHub Actions.

TECHNICAL SKILLS

AI / ML: Python (expert), scikit-learn, PyTorch, TensorFlow, LLMs, Prompt Engineering, Hugging Face

Data Engineering: SQL (advanced), PostgreSQL, ETL Pipelines, Pentaho, Airflow (concepts), Spark (concepts)

MLOps / DevOps: FastAPI, Docker, REST APIs, GitHub Actions CI/CD, MLflow, Git

Finance / Econometrics: Time Series (ARIMA, GARCH), Panel Data, Causal Inference, VaR, Backtesting

Business Intelligence: Power BI (DAX, Power Query), Excel VBA, Data Storytelling, SQL Server

Languages: Spanish (Native), English (B2 — technical reading/writing proficient)

PROFESSIONAL EXPERIENCE

Commercial Advisor — Vamoret S.A. (Palmón Group) | Nov 2022 – Present

- Lead a team of 3 people in customer service and operational improvement.
- Designed executive Power BI dashboards with real-time KPIs, RFM segmentation, and DAX forecasting — deployed in production.

- Automated monthly inventory reporting with Excel VBA macros, reducing processing time by 60%.
- Applied data analysis for pricing strategies and customer retention decisions.

Commercial Advisor — Hularuss S.A. (PepsiCo) | Oct 2018 – Mar 2021

- Managed a portfolio of clients using data-driven strategies (Excel + Power BI).
- Trained new advisors in negotiation techniques and NLP (Neuro-Linguistic Programming).
- Implemented inventory rotation and merchandising optimization at points of sale.

EDUCATION

B.Eng. Data Science & Artificial Intelligence — Universidad de Guayaquil | 2023 – Present (4th semester)

B.A. Economics — Universidad Técnica de Manabí (UTM) | 2021 – Present (7th semester, online)

★ Managing two simultaneous university degrees remotely — demonstrates exceptional time management, self-discipline, and commitment to continuous learning.

CERTIFICATIONS

- Python for Finance & Algorithmic Trading — Udemy (40h)
- SQL for Big Data — IBM (40h)
- Deep Learning Specialization — Coursera (in progress)
- LLMs & Prompt Engineering — DeepLearning.AI (in progress)
- Digital Marketing & Web Analytics — Google Activate (50h)
- Power BI for Data Analysis — Universidad de Guayaquil